Hallucination Detection from Transformer Activations

Peter Racioppo
Independent Researcher
Los Angeles, CA 90731
pcracioppo@gmail.com

March 25, 2026

Abstract

Large Language Models (LLMs) often produce fluent but incorrect responses, limiting their reliability in high-stakes settings. We propose a lightweight method for estimating answer correctness by probing internal transformer activations. We extract features from attention patterns, residual stream dynamics, and output distributions at the first generated token, and train a small neural network to predict correctness. We show that these internal signals are substantially more informative than output probabilities, indicating that model uncertainty is encoded in intermediate representations. We integrate the estimator into a Reflective Retrieval-Augmented Generation (RRAG) framework to selectively trigger retrieval and choose between zero-shot and retrieval-augmented responses. This achieves comparable or slightly improved accuracy relative to always-on RAG while reducing retrieval cost. These results suggest that internal activations provide a practical basis for confidence estimation and adaptive inference.

1 Introduction

Large Language Models (LLMs) remain prone to producing confident but incorrect responses, limiting their reliability in high-stakes applications. Retrieval-augmented generation (RAG) can improve factual accuracy by incorporating external knowledge, but introduces additional cost and does not consistently improve performance.

A central challenge is estimating when a model's output can be trusted. Common approaches rely on surface-level signals such as output probabilities or heuristic uncertainty measures, which are often poorly aligned with correctness. In contrast, recent work suggests that uncertainty may be encoded in internal transformer activations.

In this work, we show that correctness can be predicted from internal activations. We extract features from attention patterns, residual stream dynamics, and output distributions, and train a lightweight neural network to estimate correctness. This estimator can also be used for hallucination detection.

We further use this estimator to guide inference. In a Reflective RAG setting, the model first produces a zero-shot (ZS) response and triggers retrieval only when predicted correctness is low. When both ZS and RAG responses are available, the system selects the response with higher predicted correctness, yielding an adaptive strategy that balances accuracy and computational cost.

Our contributions are:

- A method for estimating answer correctness from internal transformer activations.
- Empirical evidence that internal signals are substantially more informative than output probabilities.
- An adaptive RAG framework that improves the accuracy–efficiency trade-off using learned confidence estimates.

2 Related Work

Hallucination Detection and Confidence Estimation A growing body of work studies how to detect hallucinations and estimate the reliability of language model outputs. Early approaches rely on surface-level signals such as token probabilities, entropy, or self-consistency [Kadavath et al., 2022, Manakul et al., 2023], while more recent methods incorporate external verification via retrieval [Shuster et al., 2021, Nakano et al., 2022] or auxiliary factuality benchmarks [Lin et al., 2022].

These approaches either depend on weak proxies or incur significant computational overhead. In contrast, we focus on correctness estimation from internal model states, avoiding repeated sampling and external retrieval during inference.

Internal Representations as Belief and Confidence Signals Recent work shows that transformer activations encode information about model correctness more reliably than output probabilities [Azaria and Mitchell, 2023, Burns et al., 2024]. In parallel, emerging theoretical perspectives suggest that these representations reflect structured belief states. Shai et al. [2025] show that the residual stream linearly encodes belief states over latent data-generating processes, with geometry reflecting predictive structure and long-range information about future outcomes.

These results provide a principled basis for interpreting internal activations as signals of uncertainty. Our approach builds on this view by combining attention structure, residual stream dynamics, and output features into a unified correctness estimator, and by introducing accumulated attention evidence as a scale-sensitive measure of support. Our formulation is also aligned with probabilistic interpretations of attention as precision-weighted estimation [Racioppo, 2026], where pre-softmax attention scores correspond to accumulated evidence.

Retrieval-Augmented Generation and Adaptive Inference Retrieval-Augmented Generation (RAG) improves factual accuracy by incorporating external knowledge [Lewis et al., 2021, Guu et al., 2020, Nakano et al., 2022], but introduces latency and can degrade performance when retrieval is noisy. Recent work explores adaptive retrieval strategies, triggering retrieval based on uncertainty or intermediate signals [Asai et al., 2023, Mallen et al., 2023].

Our Reflective RAG framework contributes to this line of work by using a learned correctness estimator derived from internal activations to both trigger retrieval and select between candidate responses, directly optimizing for correctness rather than relying on heuristic uncertainty measures. More broadly, our results reinforce that uncertainty is not well captured by output probabilities alone [Guo et al., 2017], and is better understood through internal representations.

3 Method

3.1 Problem Setup

Given an input prompt x, a language model produces a response y. We aim to estimate the probability that the response is correct:

$$p_\theta = P(\text{correct} \mid x, y) \tag{1}$$

We treat this as a post-hoc estimation problem, using features extracted from the model's internal activations during the forward pass.

3.2 Accumulated Attention Evidence

Attention can be interpreted as an aggregation of evidence from context tokens. In standard Softmax attention, weights are obtained by normalizing a set of unnormalized scores $\tilde{A}_{ij}$:

$$A_{ij} = \frac{\tilde{A}_{ij}}{\sum_{j'} \tilde{A}_{ij'}}.$$

This normalization enforces $\sum_j A_{ij} = 1$, removing the absolute scale of the scores and retaining only their relative proportions.

In likelihood-based formulations of attention, the unnormalized scores $\tilde{A}_{ij}$ arise from evaluating the consistency between a query and each context token under a noise model [Racioppo, 2026]. In this setting, attention corresponds to a precision-weighted aggregation of noisy observations, where each term contributes proportionally to its reliability. Under this formulation, the sum of unnormalized weights corresponds to the total precision of the aggregated estimate.

Motivated by this interpretation, we define accumulated attention evidence as:

$$p_{\text{tot},i} = \sum_j \tilde{A}_{ij}. \tag{2}$$

This quantity provides a scale-sensitive measure of internal support. Large values of $p_{\text{tot},i}$ arise when many tokens contribute strong, consistent evidence, while smaller values indicate weaker or more diffuse support.

Importantly, $p_{\text{tot},i}$ should be interpreted as a relative measure of internal agreement rather than a calibrated probability. Correlated or redundant context can lead to overcounting, so high values may arise either from diverse supporting evidence or from redundant, self-consistent patterns.

In practice, we compute a stabilized version using log-sum-exp aggregation across attention heads at selected layers. This yields a robust, scale-sensitive feature that complements normalized attention patterns and output-based uncertainty measures.

3.3 Internal Feature Extraction

For each input, we extract a feature vector $\mathbf{z}$ from the first generated token, consisting of the following four components:

Accumulated Attention Evidence Log-sum-exp aggregation of attention scores across heads at selected layers (Layers 11 and 21), capturing how strongly the model attends to relevant context.

Attention Head Magnitudes ℓ_2 norms of individual attention head outputs, reflecting their contribution to the residual stream.

Residual Stream Trajectory Layer-wise residual magnitudes and innovations (changes between successive layers), capturing the evolution of internal representations.

Output Distribution Features Top-k log probabilities, entropy, and probability margins, capturing surface-level uncertainty.

3.4 Confidence Model

We train a feedforward network $f_\theta(\mathbf{z})$ to estimate correctness:

$$p_\theta = \sigma(f_\theta(\mathbf{z})) \tag{3}$$

where σ is the sigmoid function. The model is a two-layer MLP trained with binary cross-entropy loss.

3.5 Decision Rule and Interpretation

At inference time, we compute correctness estimates for each candidate response. The estimator can be interpreted as a value function over inference strategies, selecting:

$$y^* = \arg \max_{y \in \{\text{ZS},\text{RAG}\}} P(\text{correct} \mid \mathbf{z}, \text{mode}) \tag{4}$$

A binary feature indicating whether retrieval was used allows the model to condition on the inference regime while still relying on internal signals.

3.6 Training Data

Training data is constructed by generating model outputs on benchmark datasets and labeling each response as correct or incorrect. For Reflective RAG, we include paired zero-shot and retrieval-augmented responses, enabling a unified estimator across inference modes.

We note that this formulation estimates correctness from a single decoding step; sequence-level extensions are discussed in Section 5.

4 Experiments

We evaluate two questions: (1) whether internal signals predict correctness, and (2) whether these estimates can be effectively used to adaptively trigger RAG.

4.1 Model Setup

We conduct experiments using TinyLlama-1.1B-Chat-v1.0 as the base language model. For retrieval in the RAG setting, we use all-MiniLM-L6-v2 to compute dense embeddings for query-document matching.

All features are extracted from the first generated token, allowing correctness to be estimated from a single forward pass. Responses are generated using greedy decoding with a maximum of 50 new tokens. For zero-shot prompting, we use a simple question–answer format ("Question: ... Answer:"). We use deterministic decoding to ensure that correctness signals reflect model uncertainty rather than sampling variability. In the RAG setting, retrieved context is prepended to the prompt using a standard instruction template. Generated responses are truncated to the first answer span.

4.2 Ablation Studies (LAMBADA Dataset)

We evaluate the contribution of different feature groups using k-fold cross-validation on the LAMBADA dataset [Paperno et al., 2016]. The task is to predict whether the model's generated token is correct, and performance is measured using F1-score on the hallucination class. We also report the baseline model accuracy to contextualize task difficulty.

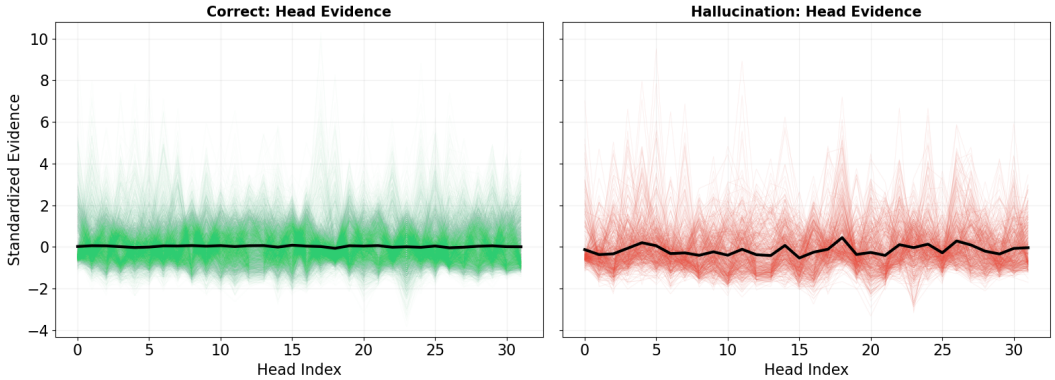
Figures 1 and 2 illustrate representative internal signals extracted at the first output token. These include attention-based evidence, head magnitudes, and residual stream dynamics, which form the basis of our feature representation.

Figure 3 compares the full model against classifiers trained on individual feature subsets. Output-based features (logits, entropy, and probability margins) exhibit minimal predictive power in isolation, confirming that standard confidence proxies are poorly aligned with correctness.

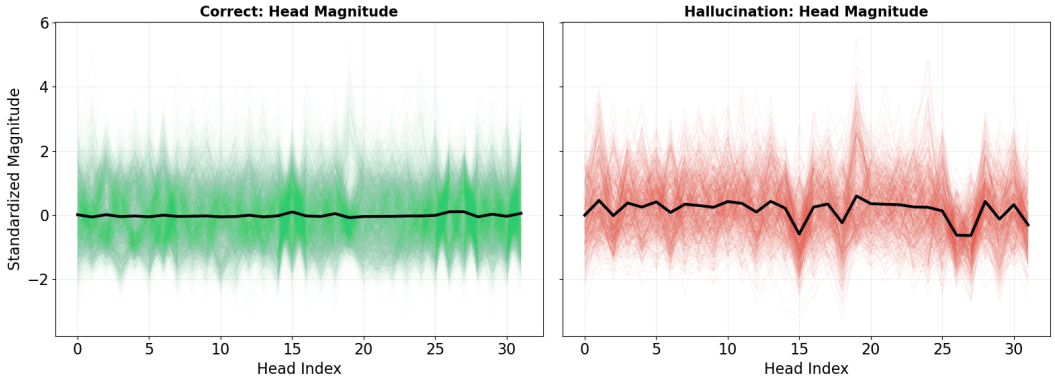
In contrast, internal transformer signals are substantially more informative. Accumulated attention evidence and attention head magnitudes are each predictive of correctness, indicating that reliable signals emerge from how information is aggregated and propagated through attention. Residual stream features also contribute, though they are weaker in isolation. Combining feature groups yields significant improvements, with attention-based evidence and head magnitudes providing the largest gains, suggesting that these signals capture complementary aspects of the model's internal computation.

We further examine the role of layer depth in Figure 4. Features from later layers are more informative than those from intermediate layers, but the best performance is achieved by combining both. This indicates that correctness signals are distributed across the network and refined through the forward pass.

While these results are demonstrated on a single model and dataset, they suggest that internal activations encode generalizable signals of correctness beyond output probabilities.



Top: Accumulated attention evidence, for correct and incorrect answers (means are shown in black).



Bottom: Attention head magnitudes across selected milestones, for correct and incorrect answers (means are shown in black).

Figure 1: Distribution of internal signals at the first output token. Both head evidence and magnitude exhibit, on average, greater volatility across heads when the model's output is incorrect.

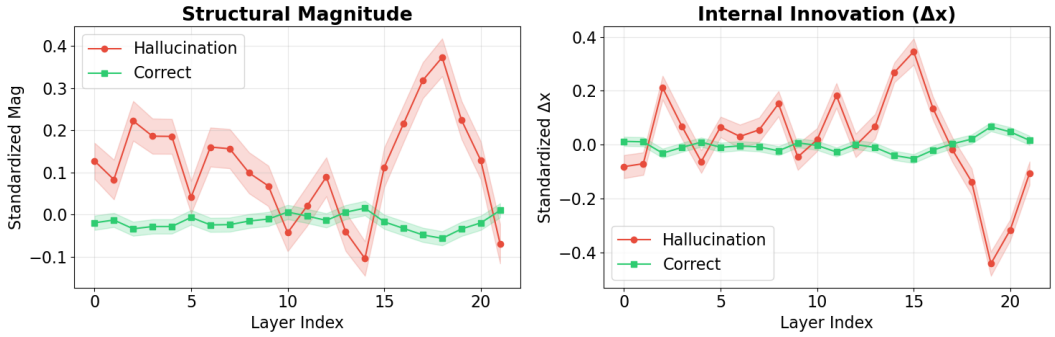


Figure 2: Trajectory of residual stream magnitudes and innovations (cosine similarity) across model layers. For incorrect answers, these values exhibit volatile oscillations.

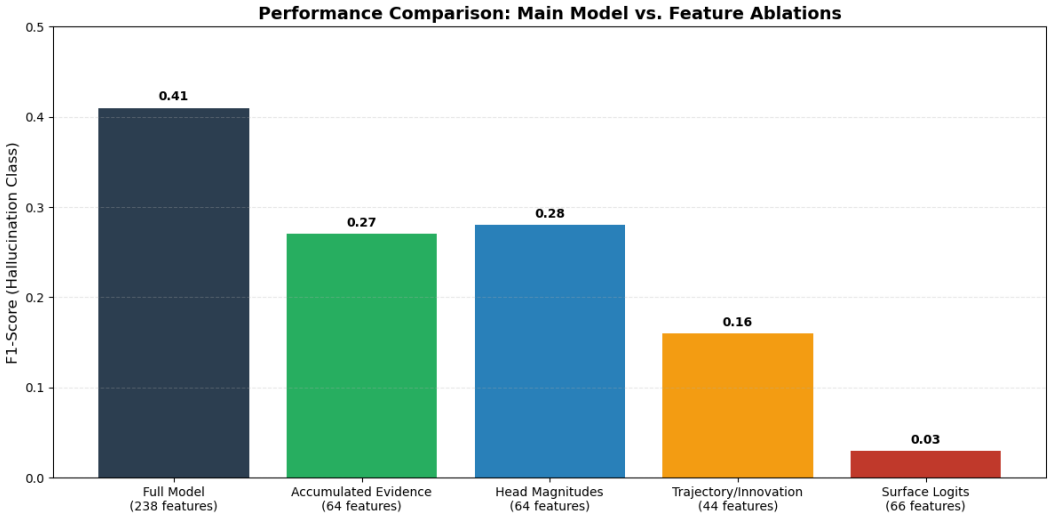


Figure 3: Performance comparison of the full classifier against individual feature category ablations.

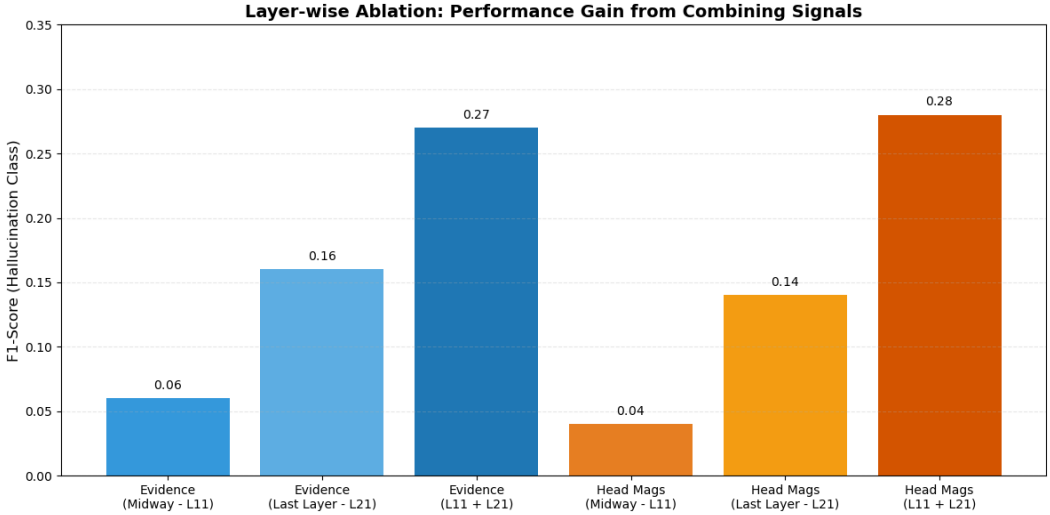


Figure 4: Incremental F1-score gains from combining features across layers.

4.3 Reflective RAG Results

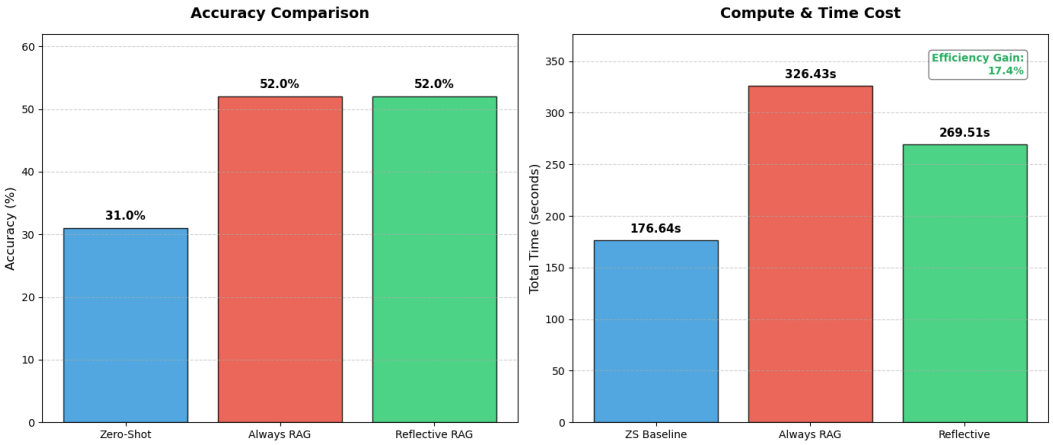
We integrate the correctness estimator into an adaptive RAG framework. Given a query, the model first produces a zero-shot (ZS) response. If the predicted probability of correctness falls below a fixed threshold, retrieval is triggered and a retrieval-augmented (RAG) response is generated. When both responses are available, the system selects the one with higher predicted correctness.

We evaluate this approach on TriviaQA [Joshi et al., 2017]. The estimator is trained on paired ZS and RAG responses, with a binary feature indicating the inference mode. We train on 4,000 queries (8,000 responses) and evaluate on 500 held-out queries. Figure 5 shows representative outcomes.

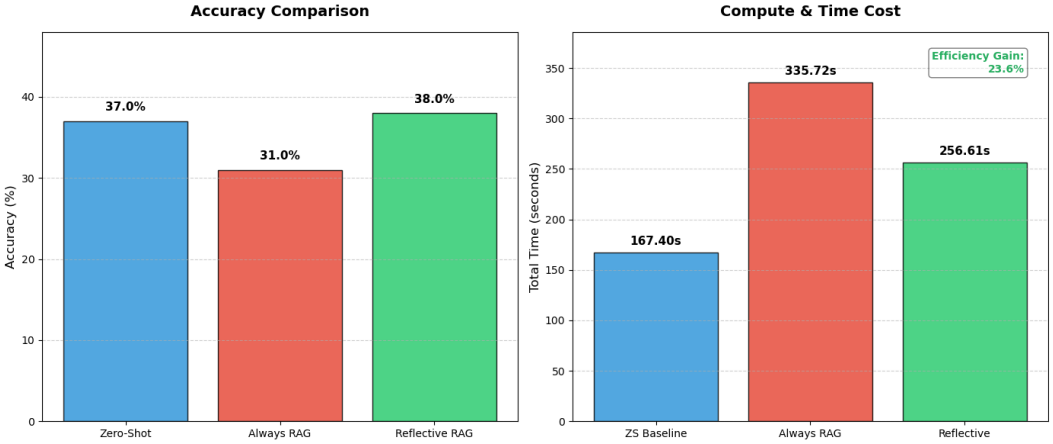
The system learns to invoke retrieval when it improves correctness and avoid it when it introduces noise, matching the benefits of RAG while mitigating its failures. These effects may be more pronounced for smaller models such as TinyLlama-1.1B, where the ability to integrate retrieved context is limited, but this setting provides a useful testbed for evaluating adaptive retrieval.

The adaptive strategy achieves performance comparable to always-on RAG while reducing retrieval usage by approximately 15–20%, yielding a favorable accuracy–efficiency trade-off. This supports

interpreting the estimator as a value function over inference strategies, selecting the response with the highest predicted correctness.



Example 1 (RAG-Helpful): Accuracy improves from 31% (ZS) to 52% (RAG). Reflective RAG matches the higher accuracy while reducing time cost by 17.44%.



Example 2 (RAG-Hurtful): Accuracy drops from 37% (ZS) to 31% (RAG). Reflective RAG achieves the highest accuracy (38%) by selectively ignoring harmful context, with a 23.56% efficiency gain.

Figure 5: System performance under different RAG conditions ($N = 100$). In both cases, Reflective RAG improves the accuracy–efficiency trade-off by selectively invoking or rejecting retrieval.

Using a decision threshold of 0.5, Reflective RAG maintains accuracy comparable to always-on RAG while reducing retrieval usage. In simulation, this yields approximately 15–20% savings in retrieval cost by skipping queries where zero-shot confidence is high.

Qualitatively, retrieval is triggered primarily when it improves correctness and avoided when it introduces noise, indicating that the learned confidence estimates align well with downstream utility.

5 Future Work: Sequence-Level Confidence Estimation

The current approach estimates correctness from a single forward pass at the first generated token, providing a snapshot of model uncertainty. A natural extension is to model confidence over the full decoding trajectory.

At each decoding step t, we extract a feature vector $\mathbf{z}_t$ and consider the sequence $\{\mathbf{z}_1, \ldots, \mathbf{z}_T\}$. Aggregating this sequence using a sequence model or pooling mechanism yields a refined estimate of correctness.

This formulation can capture dynamics not observable at a single timestep, including shifts in attention structure, degradation of residual coherence, or divergence in output distributions. It also enables finer-grained control, such as triggering retrieval during decoding rather than after an initial prediction.

More broadly, sequence-level confidence estimation provides a pathway toward real-time monitoring and correction of model outputs.

6 Conclusion

We presented a post-hoc method for estimating the correctness of language model outputs using internal transformer signals. By extracting features from attention mechanisms, residual stream dynamics, and output distributions, we train a lightweight model that predicts whether a generated response is correct.

Our results show that internal activations provide substantially stronger indicators of correctness than surface-level output probabilities.

We further demonstrate that this estimator can guide inference in a Reflective RAG framework. By selectively triggering retrieval and choosing between candidate responses based on predicted correctness, the system achieves a favorable accuracy–efficiency trade-off.

These findings suggest that internal model states provide a practical signal for post-hoc confidence estimation and motivate adaptive inference strategies in which models dynamically allocate computation based on their own uncertainty.

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